

EPT-TEST-19(IIITH,FULL LENGTH TEST-4)

Question Paper Contain

50 Aptitude | 10 Algorithms | 7 C-Programming & DS |

11 Maths | 6 CN | 5 DBMS | 8 OS | Digital 3 |

Total 100 Marks | 180 Minutes Duration

+1 Mark to each correct answer -0.25 Mark to each wrong answer

Q1.) People were prohibited_____their vehicles near the entrance of the main administrative building.

- A. to park**
- B. from parking**
- C. parking**
- D. to have parked**

Q2.) Select the word that fits the analogy:

Do : Undo :: Trust : _____

- A. Entrust**
- B. Intrust**
- C. Distrust**
- D. Untrust**

Q3.) Stock markets _____ at the news of the coup.

- A. poised**
- B. plunged**
- C. plugged**
- D. probed**

Q4.) If P, Q, R, S are four individuals, how many teams of size exceeding one can be formed, with Q as a member?

- A. 5**
- B. 6**
- C. 7**
- D. 8**

Q5.) Non-performing Assets (NPAs) of a bank in India is defined as an asset, which remains unpaid by a borrower for a certain period of time in terms of interest, principal, or both. Reserve Bank of India (RBI) has changed the definition of NPA thrice during 1993–2004, in terms of the holding period of loans. The holding period was reduced by one quarter each time. In 1993, the holding period was four quarters (360 days).

Based on the above paragraph, the holding period of loans in 2004 after the third revision was _____ days.

- A. 45**
- B. 90**
- C. 135**
- D. 180**

Q6.) Select the next element of the series: Z, WV , RQP, _____

- A. LKJI**
- B. JIHG**
- C. KJIH**
- D. NMLK**

Q7.)Build : Building :: Grow : _____

- A. Grown**
- B. Grew**
- C. Growth**
- D. Growed**

Q8.)I do not think you know the case well enough to have opinions. Having said that, I agree with your other point. What does the phrase “having said that” mean in the given text?

- A. as opposed to what I have said**
- B. despite what I have said**
- C. in addition to what I have said**
- D. contrary to what I have said**

Q9.) Crowd funding deals with mobilisation of funds for a project from a large number of people, who would be willing to invest smaller amounts through web-based platforms in the project.

Based on the above paragraph, which of the following is correct about crowd funding?

- A. Funds raised through unwilling contributions on web-based platforms**
- B. Funds raised through large contributions on web-based platforms**
- C. Funds raised through coerced contributions on web-based platforms**
- D. Funds raised through voluntary contributions on web-based platforms**

Q10.) If $x+2y=30$, then $((2y/5)+(x/3))+((x/5)+(2y/3))$ will be equal to

- A. 8**
- B. 16**
- C. 18**
- D. 20**

Q11.) The number of divisors of 6000, where 1 and 6000 are also considered as divisors of 6000 is

- A. 40**
- B. 50**
- C. 60**
- D. 30**

Q12.) The sum of the series $1/(1.2)+1/(2.3)+\dots+1/(n(n+1))+\dots$ is

- A. 1**
- B. $1/2$**
- C. 0**
- D. non-existent**

Q13.) The conditions on a , b and c under which the roots of the quadratic equation $ax^2+bx+c=0$, $a \neq 0$, $b \neq 0$ and $c \neq 0$, are unequal magnitude but of the opposite signs, are the following:

- A. a and c have the same sign while b has the opposite sign.
- B. b and c have the same sign while a has the opposite sign; or a and b have the same sign while c has the opposite sign.
- C. a and c have the same sign.
- D. a, b and c have the same sign.

Q14.) The sum of the series $3+11+\dots+(8n-5)$ is

- A. $4n^2-n$
- B. $8n^2+3n$
- C. $4n^2+4n-5$
- D. $4n^2+2$

Q15.) Find the next term in the sequence: 15O, 19S, 21U, _____?

- A. 25Y
- B. 23W
- C. 25W
- D. 23Y

Q16.) Select the word that fits the analogy:

Fuse : Fusion :: Use : _____

- A. Usage
- B. User
- C. Uses
- D. Usion

Q17.) Define $[x]$ as the greatest integer less than or equal to x , for each $x \in (-\infty, \infty)$. If $y = [x]$, then area under y for $x \in [1, 4]$ is

_____.

- A. 1
- B. 3
- C. 4
- D. 6

Q18. a, b, c are real numbers. The quadratic equation $ax^2 - bx + c = 0$ has equal roots, which is β , then

- A. $\beta = b/a$
- B. $\beta^2 = ac$
- C. $\beta^3 = bc/(2a^2)$
- D. $\beta^2 \neq 4ac$

Q19.) The sum of the first n terms in the sequence $8, 88, 888, 8888, \dots$ is _____.

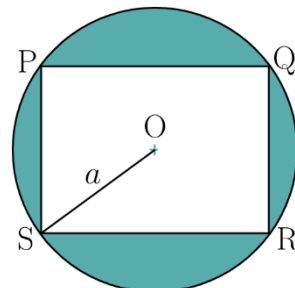
- A. $((81/80)(10^n - 1)) + (9/8)n$
- B. $((81/80)(10^n - 1)) - (9/8)n$
- C. $((80/81)(10^n - 1)) + (8/9)n$
- D. $((80/81)(10^n - 1)) - (8/9)n$

Q20.) It is quarter past three in your watch. The angle between the hour hand and the minute hand is _____.

- A. 0°
- B. 7.5°
- C. 15°
- D. 22.5°

Q21.) A circle with center O is shown in the figure. A rectangle $PQRS$ of maximum possible area is inscribed in the circle. If the radius of the circle is a , then the area of the shaded portion is _____.

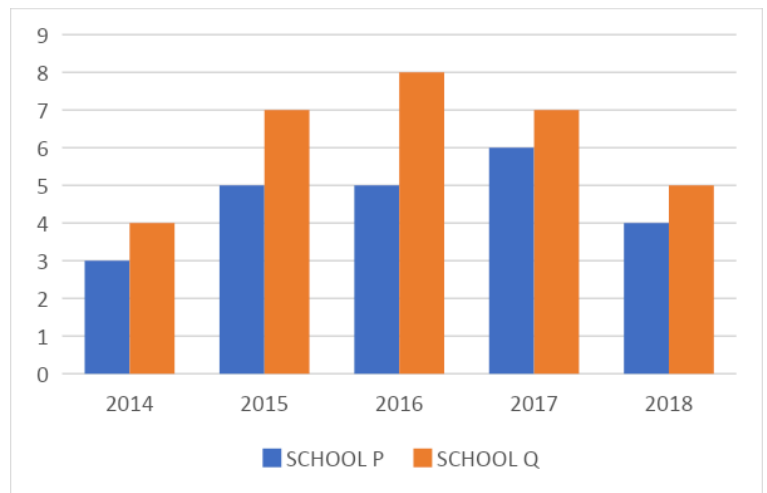
- A. $\pi a^2 - a^2$
- B. $\pi a^2 - \sqrt{2}a^2$
- C. $\pi a^2 - 2a^2$
- D. $\pi a^2 - 3a^2$



Q22.) The following figure shows the data of students enrolled in 5 years (2014 to 2018) for two schools P and Q. During this

period, the ratio of the average number of the students enrolled in school P to the average of the difference of the number of students enrolled in schools P and Q is _____.

- A. 8:23
- B. 23:8
- C. 23:31
- D. 31:23

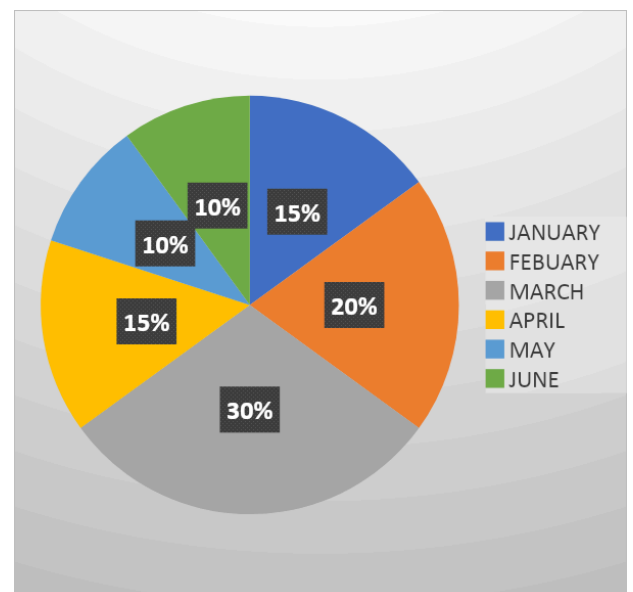


Q23.) The monthly distribution of 9-Watt LED bulbs sold by two firms X and Y from January to June 2018 is shown in the pie-chart and the corresponding table. If the total number of LED bulbs sold by the two firms during April-June 2018 is 50000, then the number of LED bulbs sold by the firm Y during April-June 2018 is _____.

Percentage of 9 Watt LED bulbs sold by the firms X and Y from January 2018 to June,2018

MONTH	Ratio of led bulb sold by 2 firms x:y
JANUARY	7:8
FEBRUARY	2:3
MARCH	2:1
APRIL	3:2
MAY	1:4
JUNE	9:11

- A. 11250
- B. 9750
- C. 8750
- D. 8250

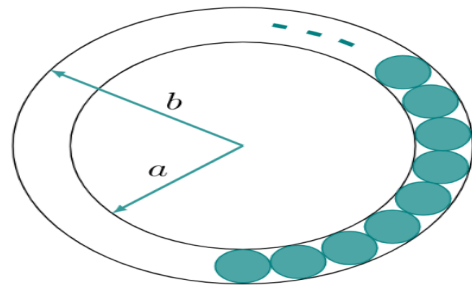


Q24.) If $P=3$, $R=27$, $T=243$, then $Q+S=$ _____

- A. 40
- B. 80
- C. 90
- D. 110

Q25.) The figure below shows an annular ring with outer and inner as b and a , respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is _____.

- A. $\pi[(b^2 - a^2) - (n/4)(b - a)^2]$
- B. $\pi [(b^2 - a^2) - n(b - a)^2]$
- C. $\pi [(b^2 - a^2) + (n/4)(b - a)^2]$
- D. $\pi [(b^2 - a^2) + n(b - a)^2]$



Q26.) The equation of any circle passing through the origin and with its centre on the X-axis is given by

- A. $x^2 + y^2 - 2ax = 0$
- B. $y^2 - 2ay + x^2 = 0$
- C. $x^2 + y^2 = a^2$
- D. $x^2 - 2ay + y^2 = 0$

Q27.) Consider a circle with center at origin and radius $2\sqrt{2}$. A square is inscribed in the circle whose sides are parallel to the X and Y axes. The coordinates of one of the vertices of this square are

- A. $(2, -2)$
- B. $(2\sqrt{2}, -2)$
- C. $(-2, 2\sqrt{2})$
- D. $(2\sqrt{2}, -2\sqrt{2})$

Q 28.) Let $y=[3245.7]$ where $[a]$ denotes the greatest integer less than or equal to a . Then

- A. $y=0$**
- B. $y=1$**
- C. $y=2$**
- D. $y=3$**

Q29.) The number of ways in which the number 1440 can be expressed as a product of two factors is equal to

HINT:-> If n is not a perfect square then no of ways it can be written as product of factors = $(\# \text{ of factors of } n)/2$

If n is a perfect square then no of ways it can be written as product of factors = $((\# \text{ of factors of } n)-1)/2$

- A. 18**
- B. 720**
- C. 360**
- D. 36**

Q30.) Let k and n be integers greater than 1. Then $(kn)!$ is not necessarily divisible by

- A. $(n!)^k$**
- B. $(k!)n$**
- C. $n! \cdot k!$**
- D. 2^{kn}**

Q31.) The 5000th term of the sequence 1,2,2,3,3,3,4,4,4,4,... is

- A. 98**
- B. 99**
- C. 100**
- D. 101**

Q32.) The total number of factors of 3528 greater than 1 but less than 3528 is

- A. 35
- B. 36
- C. 34
- D. None of these

Q33.) If 9 apples cost Rs. 100 and 10 mangoes cost Rs. 100 then the maximum number of fruits one can buy with Rs. 150 is

- A. 15
- B. 17
- C. 19
- D. 19

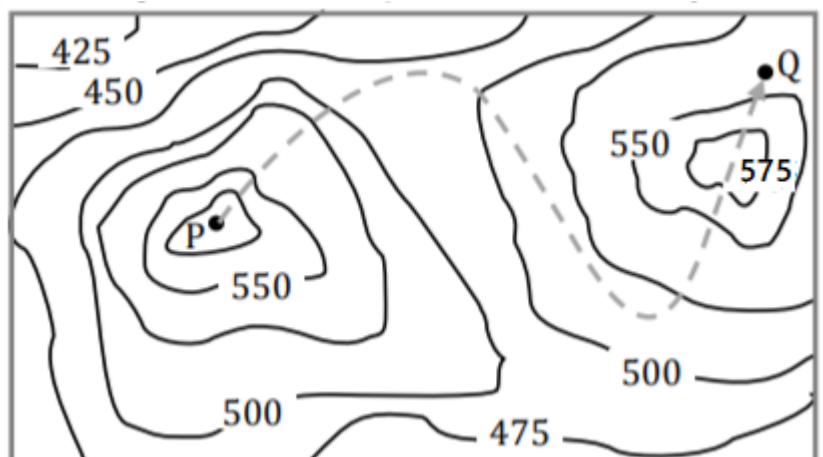
Q34.) Pick the word which fills the blank suitably
The messages went _____ until a decision was made.

- A. back and forth
- B. deliberated
- C. discussed
- D. back

Q35.) A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.

The path from P to Q is best described by

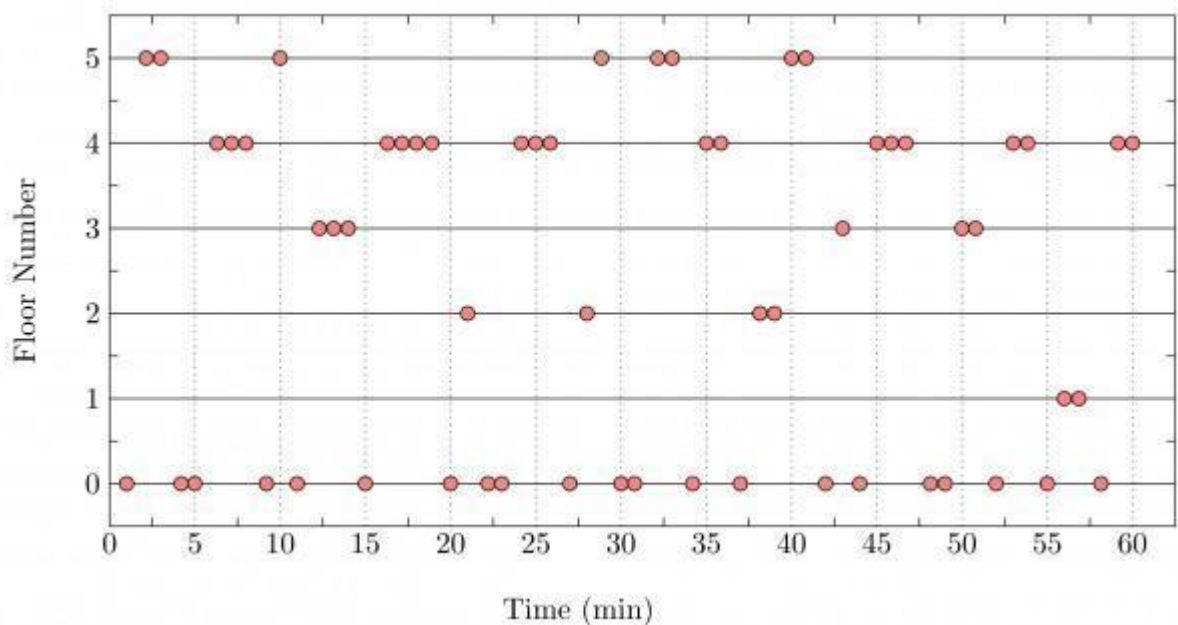
- A. Up-Down-Up-Down
- B. Down-Up-Down-Up
- C. Down-Up-Down
- D. Up-Down-Up



Q36.) The expression $(11.98 \times 11.98 + 11.98 \times a + 0.02 \times 0.02)$ will be a perfect square for 'a' equal to _____

- A. 0.02
- B. 0.2
- C. 0.04
- D. 0.4

Q 37.) The points in the graph below represent the halts of a lift for a durations of 1 minute, over a period of 1 hour.



Which of the following statements are correct?

- i.) The elevator moves directly from any non-ground floor to another non-ground floor over the one-hour period.
- ii.) The elevator stays on the fourth floor for the longest duration over the one-hour period.

- A. Only i
- B. Only ii
- C. Both i and ii
- D. Neither i nor ii

Q38.) Budhan covers a distance of 19 km in 2 hours by cycling one fourth of the time and walking the rest. The next day he cycles (at the same speed as before) for half the time and walks the rest (at the same speed as before) and covers 26 km in 2 hours. The speed in km/h at which Budhan walk is

- A. 1**
- B. 4**
- C. 5**
- D. 6**

Q39.) A map shows the elevations of Darjeeling, Gangtok, Kalimpong, Pelling, and Siliguri. Kalimpong is at a lower elevation than Gangtok, Pelling is at a lower elevation than Gangtok. Pelling is at a higher elevation than Siliguri. Darjeeling is at a higher elevation than Gangtok. Which of the following statements can be inferred from the paragraph above?

- i. Pelling is at a higher elevation than Kalimpong**
- ii. Kalimpong is at a lower elevation than Darjeeling**
- iii. Kalimpong is at a higher elevation than Siliguri**
- iv. Siliguri is at a lower elevation than Gangtok**

- A. Only ii**
- B. Only ii and iii**
- C. Only ii and iv**
- D. Only iii and iv**

Q40.) Bhaichung was observing the pattern of people entering and leaving a car service center. There was a single window where customers were being served. He saw that people inevitably came out of the center in the order that they went in. However, the time they spent inside seemed to vary a lot: some people came out in a matter of minutes while for others it took much longer.

From this, what can conclude?

- A. The center operators on a first-come-first-served basis, but with variable service times, depending on specific customer needs.**
- B. Customers were served in an arbitrary order, since they took varying amounts of time for service completion in the center.**
- C. Since some people came out within a few minutes of entering the center, the system is likely to operate on a last-come-first-served basis.**
- D. Entering the center early ensured that one would have shorter service times and most people attempted to do this.**

Q41.) What must be added to each term of the ratio 2:5 so that it may equal to 5:6?

- A. 65**
- B. 78**
- C. 13**
- D. 12**

Q42.) Find the greatest 6-digit number, which is a multiple of 12.

- A. 999980**
- B. 999990**
- C. 999984**
- D. None of these**

Q43.) Synonym: Mediocre

- A. medieval**

- B. industrial**
- C. agricultural**
- D. average**

Q44.) Axiom

- A. A statement that required proofs to get it established**
- B. A statement or proposition that is regarded as being established**
- C. A statement which is not relevant**
- D. A statement which is relevant to current situation**

Q45) Arrange the words given below in a meaningful sequence.

- 1. Poverty**
- 2. Population**
- 3. Death**
- 4. Unemployment**
- 5. Disease**

- A. 2,3,4,5,1**
- B. 3,4,2,5,1**
- C. 2,4,1,5,3**
- D. 1,2,3,4,5**

Q46.) One flies a kite with a thread 180 meter long. If the thread of the kite makes an angle of 60° with the horizontal line, then the height of the kite from the ground (assuming thread to be in straight line) is

- A. 50 meter**
- B. $90\sqrt{3}$ meter**
- C. $75\sqrt{3}$ meter**
- D. 90 meter**

Q 47.) 'A' sold an article to 'B' at a profit of '20%.' 'B' sold the same article to 'C' at a loss of 25% and 'C' sold the same article

to 'D' at a profit of 40%. If 'D' paid Rs. 252 for the article, then find how much did 'A' pay for it?

- A. Rs. 175
- B. Rs. 200
- C. Rs. 180
- D. Rs. 210

Q 48.) 60 percent of first number is equal to 40 percent of the second number. What is the respective ratio of the first number to the second number?

- A. 2:3
- B. 21:3
- C. 7:10
- D. Can't be determined

Q49.) The main statement is followed by four statements labeled I, II, III and IV. Choose the ordered pair of statements, where the first statement implies the second and the two statements are logically consistent with the main statement. The exam is either GATE or PGEE.

- I. The exam is GATE.
- II. The exam is not GATE.
- III. The exam is PGEE.
- IV. The exam is not PGEE.

- A I, II
- B III, IV
- C II, I
- D II, III

Q50.) What is the probability of throwing a number greater than 2 with a fair dice?

A 2/3

B 2/5

C 1

D 3/5

Q51.) Suppose that you implement a queue using a null-terminated singly-linked list, maintaining a reference to the item least recently added (the front of the list) but not maintaining a reference to the item most recently added (the end of the list). What are the worst case running times for enqueue and dequeue?

A. constant time for both enqueue and dequeue

B. constant time for enqueue and linear time for dequeue

C. linear time for enqueue and constant time for dequeue

D. linear time for both enqueue and dequeue

Q52.) Consider a Functional dependency $X \rightarrow A$ with respect to relation R. We know that there is an attribute B belonging to X such that B is not prime.

A R is not in 2NF

B R is in 2NF, but might not be in 3NF

C R is in 3NF, but might not be in BCNF

D R is in 2NF, 3NF, and in BCNF

Q53.) Let E and F be two independent events, with $P(E) = P(F) = 0.25$. Let G be another event on the same sample space, which is disjoint from both E and F, and with $P(G) = 0.25$. Then, $P(G|F)$ is

A. 11

B. 0.5

C. 0.25

D. 0

Q54.) Let $P(A)$ denote the power set of A. If $P(A) \subseteq B$ Then

- A. $2|A| \leq |B|$
- B. $2|A| \geq |B|$
- C. $2|A| < |B|$
- D. $|A| + 2 \leq |B|$

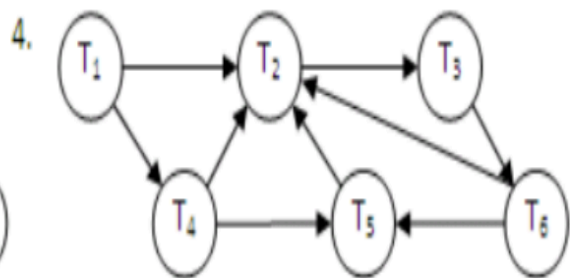
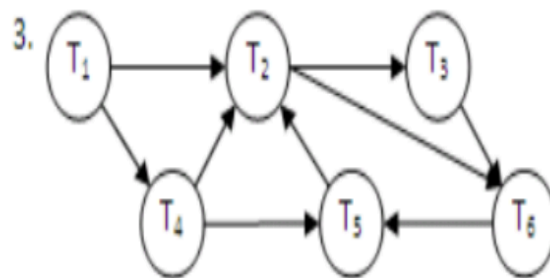
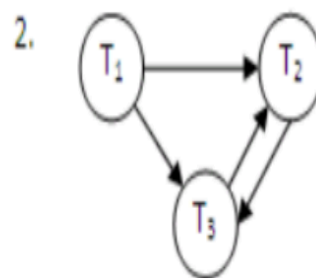
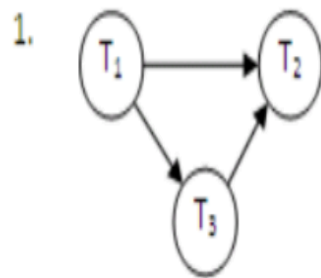
Q 55.) Match the logic gates in Column A with their equivalents in Column B.

	Column A	Column B
A. P-2, Q-4, R-1, S-3	P. 	1. 
B. P-4, Q-2, R-1, S-3	Q. 	2. 
C. P-2, Q-4, R-3, S-1	R. 	3. 
D. P-4, Q-2, R-3, S-1	S. 	4. 

Q56)A simple way to detect a state of deadlock is to construct a wait-for graph. One node is created in the graph for each transaction that is currently executing in the schedule.

Whenever a transaction T_i is waiting to lock an item X that is currently locked by a transaction T_j , it creates a directed edge $(T_i \# T_j)$. When T_j releases the lock(s) on the items that T_i was waiting for, the directed edge is dropped from the waiting-for graph.

Given the graph below, identify the deadlock situations.



Which of the above wait-for graph represents a deadlock situation?

- A. Only 3,4
- B. Only 1
- C. Only 2,3,4
- D. ALL

Q 57.) When a user submits a query, a search engine does the following. For every webpage that has been visited by the search engine, it computes a score indicating how relevant that page is to the query. Finally, it reports the pages with the

top k scores on the screen, for a number k specified by the user. A good data structure for accumulating the scores and ranking them is:

- A. a queue
- B. a heap
- C. a stack
- D. a binary search tree

Q58) Suppose we have a set of n value. There are always at least one negative value and at least one positive value in the set. What is the worst-case time complexity to find FIRST (i.e. first value from left) negative value of set?

- A. If values are stored in an array and arranged in ascending order then it takes $\theta(n)$ in worst case.
- B. If values are stored in an array and arranged in descending order then it takes $\theta(n)$ in worst case.
- C. If values are stored in a linked list (only head is known) and arranged in ascending order then it takes $\theta(n)$ in worst case.
- D. If values are stored in a linked list (only head is known) and arranged in descending order then it takes $\theta(n)$ in worst case.

59.) Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n th vertex in this BFS traversal, then the maximum possible value of n is _____

- A. 30
- B. 31
- C. 32
- D. 33

Q 60.) Suppose $g(x)$ is a polynomial function such that $g(-1)=4$ and $g(2)=7$. Then there is a number c between -1 and 2 such that

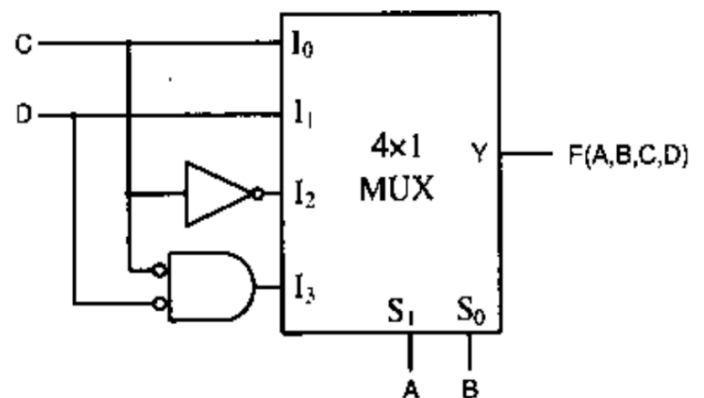
- A. $g(c)=1$
- B. $g(c)=0$
- C. $g'(c)=1$
- D. $g'(c)=0$

Q 61.) If $f(x)=x^{100}+x^{99}+\dots x+1$ then $f'(1)$ is equal to

- A. 5050
- B. 5049
- C. 50051
- D. 5656

Q 62.) The Boolean function realized by the logic circuit shown is

- A. $F=\sum m(0,1,3,5,9,10,14)$
- B. $F=\sum m(2,3,5,7,8,12,13)$
- C. $F=\sum m(1,2,4,5,11,14,15)$
- D. $F=\sum m(2,3,5,7,8,9,12)$



Q 63.) Consider a system running ten I/O-bound tasks and one CPU-bound task. Assume that the I/O bound tasks issue an I/O operation once for every millisecond of CPU computing and that each I/O operation takes 10 milliseconds to complete. Also, assume that the context-switching overhead is 0.1 millisecond and all processes are long-running tasks. What is the approx CPU utilization for a round-robin scheduler when The time quantum is 10 milliseconds?

- A. 94%
- B. 91%
- C. 90%
- D. 96%

Q64.) A digital transmission system uses Hamming code for transmitting data over a noisy channel. Suppose each transmitted bit has probability p of being received in error.

For each codeword transmission, the receiver performs minimum Hamming distance decoding and correctly decodes the message bits if and only if the channel introduces at most one-bit error.

Let the number of data bits =4 and the number of parity bits =3 hence total codeword bits =7 For $p=0.1$, the probability that a transmitted codeword is decoded correctly is (rounded off to two decimal places).

- A. 0.85**
- B. 0.72**
- C. 0.8**
- D. 0.9**

Q65.) Assume that S is a stack and $Q1$ and $Q2$ are two Queues which support the Enqueue and Dequeue operations. Consider the following pseudo code for implementing the Pop and Push operation on S .

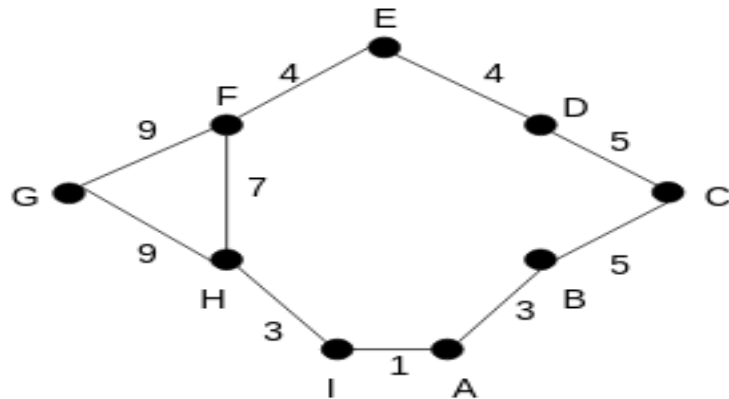
```
Push( $S,x$ )  
    A( $Q2,x$ )  
    while ( $Q1$  not empty)  
        B( $Q2,C(Q1)$ );  
    Swap( $Q1,Q2$ )  
Pop( $S$ )  
    return(D( $Q1$ ))
```

Which of the following options for the functions A , B , C , and D would correspond to correctly implementing the Push and Pop operations on the stack S ?

- A. A , B - Enqueue C , D - Dequeue**
- B. A , C - Enqueue B , D - Dequeue**
- C. A , C - Dequeue B , D - Enqueue**
- D. A , D - Enqueue B , C - Dequeue**

Q66.) Consider the following graph, the number of MST possible for the following graph is _____

- A. 2
- B. 4
- C. 9
- D. 8



Q67.) What will be the maximum data rate (in Mbps) if the bandwidth of the medium is 100 Mbps, using the selective repeat protocol, the window size is the maximum possible with the sequence number is 1024. The propagation delay between the link is 20 milliseconds. The data size is 100 bytes. [Write the floor value of the answer]

- A. 12
- B. 10
- C. 9
- D. 8

Q68.) #include<stdio.h>

enum day {Sun , Mon = -6, Tue , Wed ,Thu=4 , Fri = 8, Sat}; int

main()

{

printf("%d",Sun+Mon+Tue+Wed+Thu+Fri+Sat); return O;

}

- A. 6
- B. 7
- C. 8
- D. 9

Q69.) It's given that the size of the block is two Kilo Bytes and the size of the disk given is 112 Giga Byte, we are asked to use the bit-map to record the data blocks that are free. A Bit-map or Bit Vector is a series or collection of bits where each bit corresponds to a disk block. So, the bitmap has a size of :

- A 4MB**
- B 5MB**
- C 10MB**
- D None of the above**

Q70.) Time required to merge two singly linked lists such that the last element of the first list is followed by the first element of the second list.

- A $O(n)$**
- B $O(n^3)$**
- C $O(1)$**
- D None of the above.**

Q71.) The probability that a certain beginner at golf gets a good shot with a good club is $\frac{1}{3}$, and the probability that he gets a good shot with a bad club is $\frac{1}{4}$ and if there are 5 clubs in total and one of them is a good one and the club is chosen by the player at random, which of the following is probability that the shot is good.

- A $\frac{1}{12}$**
- B $\frac{1}{3}$**
- C $\frac{4}{15}$**
- D $\frac{7}{12}$**

Q72.) If a student is required to pass all the 6 subjects to pass in his class 10 board examination, the number of ways in which he will not be able to clear the exam is ____.

- A. 31**
- B. 63**
- C. 127**
- D. 255**

Q73.) Consider the following items with their associated profits and weights respectively, if items are to be allowed to be taken in fractions as well then, the optimal profit is given by ____. Consider the max weight of the knapsack 10 units.

- A. 23**
- B. 22**
- C. 21**
- D. 20**

ITEM	PROFIT	WEIGHT
1	5	1
2	6	2
3	4	2
4	8	1

Q74.) Consider the following frequency of characters in a text corpus

character	frequency
a	12%
b	20%
c	58%

The following string is to be encoded using Huffman encoding, how many bits would be required ____

“aaaabbbbbbc”

- A. 20**
- B. 21**
- C. 22**
- D. 23**

Q75.)
$$\int_0^{\pi} \sin x \cos^4 x \, dx$$

- A 0**
- B $\pi/4$**
- C $2/5$**
- D $1/4$**

Q76.) Which of the following is an appropriate word for the blank in the below given statement:

An Alternative method to establish that a set of attributes X, forms a key of relation R, is by checking if R is equal to the _____ of the set X.

- A Attributes**
- B Functional Dependencies**
- C Closure**
- D Prime attributes**

Q77.)

index	0	1	2	3	4	5	6	7	8	9
Array element	12	11	13	45	21	30	25	24	17	10

A randomized version of quicksort makes use of a random number generator which generates the sequence of the pivot indexes to be chosen, which of the following doesn't correspond to the worst case of quicksort.

- A 9, 1, 0, 2, 8, 4, 7, 6, 5, 3**
- B 3, 5, 6, 7, 4, 8, 2, 0, 1, 9**
- C 3, 9, 5, 1, 6, 0, 7, 2, 4, 8**
- D 9, 1, 8, 6, 3, 0, 2, 5, 7, 4**

Q78.) Let $S = \{a, b, c\}$ and $T = \{1,2,3\}$. Function F is defined as $F : S \rightarrow T$

Consider the two functions as :

- I. $F = \{(a, 3), (b, 2), (c, 1)\}$**
- II. $F = \{(a, 2), (b, 1), (c, 1)\}$**

Which of the following is correct?

- A F^{-1} exists for I, but doesn't exist for II**
- B F^{-1} exists for II, but doesn't exist for I**
- C F^{-1} doesn't exist for both I and II**
- D F^{-1} exists for both I and II**

Q79.) Let f and g be two functions given by :

$f = \{(2,4), (5,6), (8,-1), (10,-3)\}$

$g = \{(2,5), (7,1), (8,4), (10,13), (11,-5)\}$ then domain of $f + g$ is

Which of the following is correct?

A $\{2, 7, 10\}$

B $\{2, 7, 8\}$

C $\{2, 8, 11\}$

D $\{2, 8, 10\}$

Q80.) The odds against A solving a problem are 3 to 2 and the odds in favor of B solving the same problem are 5 to 4. Which of the following is the probability that the problem will be solved if both of them try the problem

A $16/21$

B $52/77$

C $11/15$

D $1/5$

Q81.) The number of longest common subsequences among the following strings is ____.

“hahahhahah”

“ahahaaha”

A. 6

B. 5

C. 4

D. 3

```

Q82.) void fork0()
{
    int pid = fork ();
    if (pid == 0)
        printf ("Hello from Child!\n");
    else
        printf ("Hello from parent!\n");
}
main ()
{
    fork0();
}

```

Which of the following is correct?

- A The order of execution is deterministic
- B Parent and child executes concurrently
- C The program will not execute
- D None of the above

Q83.) Consider a swapping system in which memory consists of the following hole sizes in memory order: 10KB, 4KB, 20KB, 18KB, 7KB, 9KB, 12KB, and 15KB. Which hole size is taken for the third segment request out of the given three successive segment requests using First Fit?

Request 1. 12KB

Request 2. 10KB

Request 3. 8KB

A 20KB

B 18KB

C 10KB

D 9KB

Q84.) Which of the following is true about TCP and UDP ?

- A In UDP header checksum is of 16 bits whereas in the TCP checksum is of size 8 bits.**
- B UDP provides the flow control but not the congestion control.**
- C TCP can be connection-oriented or connection less at different times.**
- D TCP is reliable as it guarantees delivery of data to the destination router.**

QQ85.) Which of the following options is e true ?

- A DNS is used to get the private IP address from the server.**
- B The port number of the DHCP is 53.**
- C In the Non- persistent HTTP for every object individual TCP connection is required.**
- D SMTP and POP both are used to push the mail on the server.**

Q86.) How long it takes to send a file of 640,000 bits from Host A to Host B over a circuit-switched network. Suppose that all links in the network use TDM with 24 slots and have a bit rate of 1.536 Mbps. Also suppose that it takes 500 msec to establish an end-to-end circuit before Host A can begin to transmit the file. How long(in sec) does it take to send the file?

- A. 11.5**
- B. 10.5**
- C. 12.5**
- D. 9.5**

Q87.) How much time is required to transmit the HTML base file and six objects using the non-persistent HTTP connection. The given round trip time is 20 msec.

- A 280 msec**
- B 200 msec**
- C 240 msec**
- D 160 msec**

Q88.) The total number of combinations of n different things taken one or more at a time is 127 then the number of items is

-
- A. 6
 - B. 7
 - C. 8
 - D. 5

Q89.) 89.) Consider the following table for process details, resource allocation and maximum resource requirement

The number of processes that are able to finish is

- A. 5
- B. 3
- C. 4
- D. 2

Process	Allocation	Max
	A B C	A B C
P0	3 2 2	5 6 2
P1	0 1 0	2 1 0
P2	1 1 1	5 6 1
P3	3 2 3	9 9 6
P4	2 1 1	3 2 1

Available
A B C
2 0 0

Q90.) A channel has a bit rate of 4 Kbps and a propagation delay of 20 msec. For what frame size (in bits) does stop-and-wait give an efficiency of at least 50 percent?

- A. 150
- B. 140
- C. 130
- D. 160

Q91.) The tightest upper bound for the worst case time complexity for constructing a balanced BST when given a sorted array of integers.

- A $O(n)$
- B $O(n \log n)$
- C $O(\log n)$
- D None of the above

Q92) The number of literals obtained on simplifying the $F(P, Q, R, S, T, U) = PRT + QS'T' + PST(R + T'U) (QR' + SU)$ are _____

- A. 3
- B. 4
- C. 5
- D. 6

Q93.) CONSIDER THE following process in a system

Process id	Arrival time	Burst time	priority
P0	0	3	MEDIUM
P1	1	4	HIGHEST
P2	3	2	LOWEST

NOTE ASUUME CONTEXT SWITCHING TIME IS NEGLIGIBLE
WHICH OF THE FOLLOWING IS THE WAITING TIME FOR
PROCESS P0

- A 0
- B 2
- C 4
- D 8

Q94.) Consider the following program snippet :

```
if ( fork() || fork() )
{
    if ( !fork() )
    {
        printf("\nHi")
    }
}
printf("\nexampreptool");
```

The number of times "Hi" and "exampreptool" will get printed
is :

- A 3 and 5
- B 2 and 6
- C 2 and 5
- D 3 and 7

**Q95.) #include<stdio.h>
#include<string.h>**

```
int main()  
{  
char a[6] = "DECENT" ;  
char *ptr ="ECENT" ;  
ptr=a;  
strcpy(a, ptr);  
printf("%c", (*ptr+1));  
return 0;  
}
```

A D

B E

C C

D None of the above

Q96.) What is the output of the following c-code?

#include<stdio. h>

```
int main()  
{  
    const char *a = "SLOPE".,  
    char string[] = "ROUGH";  
    a = string;  
    while(*a)  
        printf("%c", *a++);  
    return 0;  
}
```

A ROUGH

B OUGH

C R

D Error

Q 97.)

```
#include<stdio.h>
int foo(int *arr, int n)
{
    if (n <= 0)
        return 0;
    else if (*arr%10 == 0)
        return *(arr)+foo(arr+1, n-1);
    else
        return *(arr+1) - foo(arr+1, n-1);
}
```

```
int main()
{
    int arr[] = {10,2,30,4,50};
    printf("%d", foo(arr,5));
    return 0;
}
```

- A. 8
- B. 9
- C. 7
- D. 10

Q98.)A functional dependency **F:** $X \rightarrow Y$ is termed as a useful functional dependency if and only if it satisfies all the following three conditions:

- X is not the empty set.
- Y is not the empty set.
- Intersection of X and Y is the empty set.

For a relation R with 4 attributes, the total number of possible useful functional dependencies is _____.

- A. 40
- B. 50
- C. 60
- D. 70

Q99.) Division operator in relational algebra is appropriate to handle which of the following query types?

- A. employees work on at most one of the critical projects**
- B. employees work on any one of the critical projects**
- C. employees work on at least one of the critical projects**
- D. employees work on all the critical projects**

Q100.) In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?

- A. The weak entity set MUST have total participation in the identifying relationship**
- B. The owner entity set MUST have total participation in the identifying relationship**
- C. Both weak and owner entity sets MUST have total participation in the identifying relationship**
- D. Neither weak entity set nor owner entity set MUST have total participation in the identifying relationship**

1.) SOLUTION (B)

“prohibited” means “banned” or “stopped” and should be followed by “from”.

2.) SOLUTION (C)

this is analogy question – a comparison of two otherwise unlike things based on resemblance of a particular aspect

Do is verb and Undo is used as its opposite term

Trust opposite is distrust. (or mistrust)

Entrust – put (something) into someone's care or protection.

Intrust – archaic spelling of entrust.

Untrust (wrong word)

3.) SOLUTION (B).

poised – having a composed and self-assured manner. Having self-confidence.

4.) SOLUTION (C)

Number of Individuals =4, team Size greater than 1, Q is premium member.

__ _ Q. (max three positions left to be filled and three members left: P, R, S)

By raw counting we can get 7:{PQ,RQ,SQ,PRQ,PSQ,RSQ,PQRS }.

Alternatively, we can do as follows:

As four individuals are defined distinct, and we just have to select them for the three remaining positions.

Selecting 1 person out of three, selecting 2 persons out of three and selecting all 3 persons, we get

$$3C_1 + 3C_2 + 3C_3$$

$$= 3 + 3 + 1 = 7$$

5.) SOLUTION(B).

As question says – holding period was reduced by one quarter each time. In 2004 – third revision.

In 1993 holding period – 360 days (four quarters). One quarter
–= $1/4 \times 360 = 90$.

[quarter – each of four equal or corresponding parts into which something is or can be divided.]

First Time = $360 - 90 = 270$ (reduced by one quarter)

Second Time = $270 - 90 = 180$

Third Time = $180 - 90 = 90$

6.) SOLUTION(C)

In first place we have one letter, in second place we have two-letters and so on.

Now, we can follow the pattern.

- $Z \rightarrow -3W \rightarrow -5 \rightarrow R - 7K$
- $V \rightarrow -5Q \rightarrow -7J$
- $P \rightarrow -7J$
- H

So, the correct answer is (C).

7.) SOLUTION(C)

“building” is the “noun” related to the verb “build”.

Similarly, “growth” is the noun related to the verb “grow”.

Correct Answer: C.

8.) SOLUTION B

I do not think you know the case well enough to have opinions.

- **So, I'm not expected to agree with your opinions.**

Having said that, I agree with your other point.

- **So, "having said that" is "not contradicting" but "adding something extra" which is "not in agreement" with the previous statement. "despite what I have said" is the correct meaning.**

Correct Answer: B.

9.) Solution(D)

A. Funds raised through unwilling contributions on web-based platforms

"who would be willing to invest": so False.

B. Funds raised through large contributions on web-based platforms

"invest smaller amounts": so False

C. Funds raised through coerced contributions on web-based platforms

Coerced means "forced". So, similar to option A this must be False.

D. Funds raised through voluntary contributions on web-based platforms

"willing to invest smaller amounts through web-based platforms": True.

10.) SOLUTION (B)

$$(2y/5+x/3)+(x/5+2y/3)$$

Taking LCM we get:

$$((6y+5x+3x+10y)/15)$$

$$((8x+16y)/15)$$

$$(8(x+2y)/15)=(8 \times 30)/15=16$$

Option B) is correct

11.) SOLUTION(A)

6000 can be written as $2^4 * 5^3 * 3$.

therefore any divisor of 6000 can be formed by taking any combination of the above factors.

so for '2' there are 5 choices whether to include it or not in the divisor of 6000.

similarly for '5' there are 4 choices and for '3' there are 2 choices.

therefore total no of choices = $5*4*2 = 40$.

there (A) is the correct answer.

12.)SOLUTION (A)

$$1/(1.2)+1/(2.3)+...+1/(n(n+1))+...$$

$$\begin{aligned} &\propto \\ &= \sum_{k=1}^{\infty} = 1/(k(k+1)) \end{aligned}$$

$$\begin{aligned} &\propto \\ &= \sum_{k=1}^{\infty} = (1/k) - (1/k+1) \end{aligned}$$

$$=(1-1/2+1/2-1/3+1/3-1/4)$$

$$=1-1/\infty$$

$$=1$$

13.) SOLUTION(B)

$$ax^2+bx+c=0 \Rightarrow x^2+(b/a)x+(c/a)=0; [\because a \neq 0]$$

Let α, β be the two roots of $x^2+(b/a)x+c/a=0$.

$$\therefore (x-\alpha)(x-\beta) \equiv x^2+(b/a)x+(c/a)=0$$

$$\Leftrightarrow x^2-(\alpha+\beta)x+\alpha\beta \equiv x^2+(b/a)x+(c/a) \Leftrightarrow b/a = -(\alpha + \beta), c/a = \alpha\beta$$

Since the roots are unequal and of opposite sign, their product must be negative.

$$\text{So } \alpha\beta < 0 \Rightarrow c/a < 0 \Rightarrow c \cdot (1/a) < 0$$

It means c and a must be of opposite sign. Only the option B satisfies this statement.

So the correct answer is B.

14.) SOLUTION (A)

Explanation:

The above series forms an AP

where, $a=3, d=8, a_n=(8n-5)$

So, sum of n terms of AP is given by:

$$S_n = n/2[a + a_n] = n/2[3 + 8n - 5] = n/2[8n - 2] = n(4n - 1)$$

$$\therefore S_n = 4n^2 - n$$

$\therefore A$ is the right option.

15.) SOLUTION(A)

13,17,19,23, all are prime numbers, and our pattern is prime number plus two. So it will be 15,19,21,25.

- 15th alphabet M
- 19th alphabet Q
- 21th alphabet S
- 25th alphabet Y

So, the correct answer is (A).

16.) SOLUTION(A)

Fusion is the noun form of Fuse.

Usage is the noun form of Use.

Correct option: A

17.) SOLUTION(D)

The graph
of $y=[x]$ for $x \in [1,4]$.

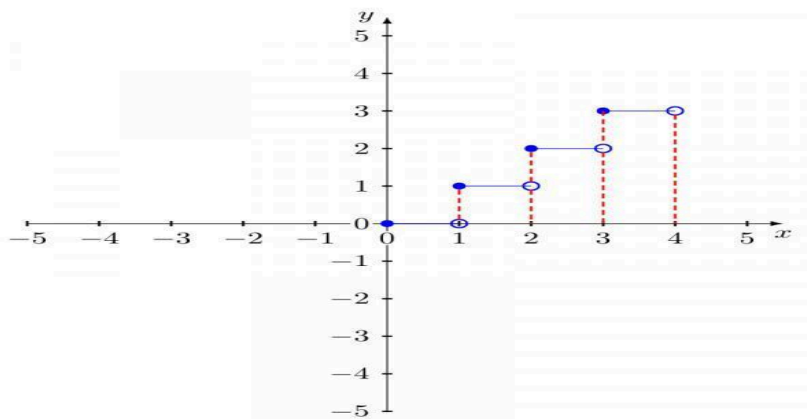
The required

$$\text{area} = (1 \times 1) + (1 \times 2) + (1 \times 3)$$

$$= 1 + 2 + 3 = 6 \text{ unit}^2.$$

[$\because$ Area of rectangle =

Base $\times$ Height]



18.) SOLUTION C

So, according to the given quadratic equation, we have

- $2\beta = b/a \rightarrow (1)$ and $\beta^2 = c/a \rightarrow (2)$

Multiplying equations (1) and (2) and simplifying, we get

$$\beta^3 = bc/2a^2$$

Option C is correct

19.) SOLUTION(D)

Let $S = 8 + 88 + 888 + \dots + 888 \dots 8$ n times

$$\Rightarrow S = 8[1 + 11 + 111 + \dots + 111 \dots 1] \text{ } n \text{ times}$$

$$\Rightarrow S = \frac{8}{9} [9 + 99 + 999 + \dots + 999 \dots 9] \text{ } n \text{ times}$$

$$\Rightarrow S = \frac{8}{9} [(10 - 1) + (10^2 - 1) + (10^3 - 1) + \dots + (10^n - 1)]$$

$$\Rightarrow S = \frac{8}{9} [10^1 + 10^2 + 10^3 + \dots + 10^n - n]$$

$$\Rightarrow S = \frac{8}{9} [(10(10^n - 1)/10 - 1) - n]$$

$$\Rightarrow S = \frac{8}{9} [(10^{n+1} - 10 - 9n)/9]$$

$$\Rightarrow S = \frac{8}{9} [10^{n+1} - 9n - 10]$$

$$\Rightarrow S = \frac{8}{9} [10^n \cdot 10 - 9n - 10]$$

$$\Rightarrow S = \left(\frac{80}{81}\right)(10^n - 1) - \frac{8}{9} n$$

So, the correct answer is D

20.) SOLUTION (B)

The angle between hour hand and minute hand of a clock at H hours and M minutes is given by

$$\Theta = 30H - \frac{11M}{2} \text{ So, at 03:15 we get their difference as } 30 \times 3 - \frac{11 \times 15}{2} = 90 - 82.5 = 7.5^\circ$$

Correct Option: B

21.) SOLUTION (C)

Theorem: "A perpendicular from the Centre of a circle to a chord, bisects the chord."

In this case, if a perpendicular is drawn from the center of the circle, it will bisect the base side of the rectangle.

The half-length of this chord would be: $a \times \cos 45^\circ = a^2$.

Hence the length of the rectangle would be $2 \times a / \sqrt{2} = a\sqrt{2}$

Area of circle : πa^2 and the area of rectangle : $2a^2$

$\therefore$ Area of remaining portion would be : $\pi a^2 - 2a^2$

22.) SOLUTION (B)

We can calculate the number of students enrolled in both schools during the given period of time,

- The total number of students enrolled in school P (in 5 years) = $(3+5+5+6+4) \times 1000 = 23000$
- Average number of students enrolled in school P = $23000/5$
- And, the total number of students enrolled in school Q (in 5 years) = $(4+7+8+7+5) \times 1000 = 31000$
- Average number of students enrolled in school Q = $31000/5$
- Average of the difference of the number of students enrolled in school P and Q = $31000/5 - 23000/5 = 8000/5$

Now the required ratio = $\frac{\frac{23000}{5}}{\frac{8000}{5}} = \frac{23}{8}$.

$\therefore$ The ratio of the average number of the students enrolled in

school P to the average of the difference of the number of students enrolled in schools P and Q is 23:8

23.) SOLUTION (B)

April 3:2 15% $5x = 50000 \cdot (15/100)$, $x = 1500$, $Y = 2x = 3000$

May 1:4 10% $5x = 50000 \cdot (10/100)$, $x = 1000$, $Y = 4x = 4000$

June 9:11 10% $20x = 50000 \cdot (10/100)$, $x = 250$, $Y = 20x = 2750$

Total bulbs sold by Y firm from April – June is $= 3000 + 4000 + 2750 = 9750$

24.) SOLUTION (C)

We can interpret the given order as:

- $P \rightarrow 31=3$
- $Q \rightarrow 32=9$
- $R \rightarrow 33=27$
- $S \rightarrow 34=81$
- $T \rightarrow 35=243$

$Q+S \rightarrow 9+81=90$

25.) SOLUTION (A)

We need to find the area of the ring first.

Area of Ring = Area of Outer Circle – Area of Inner Circle
 $= \pi b^2 - \pi a^2 = \pi (b^2 - a^2)$

Now From this ring area we have to subtract the small 'n' circles

Area of n Circles $= (\pi d^2/4) \cdot n = (\pi (b-a)^2/4) \cdot n$

The Required Unpainted Area = Ring Area – Area of Small 'n' such Circles

$= \pi (b^2 - a^2) - (\pi d^2/4) \cdot n$

$= \pi [(b^2 - a^2) - n/4(b-a)^2]$

26.) SOLUTION (A)

If a circle passes through the origin and the center lies on the x axis then the length of the abscissa will be equal to the circle's radius and the ordinate(y-coordinate) will be 0.

So, the equation of the circle will be of the form:

$$(x-a)^2+y^2=a^2$$

$$\Rightarrow x^2+y^2-2ax=0$$

$\therefore$ A is the right option.

27.) SOLUTION (A)

Let radius of the circle be r, then: $r=2\sqrt{2}$

Draw perpendiculars from this point to x and y axes respectively and let them be a and b.

Now,

$$a^2+b^2=r^2 \Rightarrow 2a^2=r^2$$

$$\Rightarrow 2a^2=(2\sqrt{2})^2=4 \times 2 \Rightarrow a=\pm 2$$

So, the coordinates are (2,-2)

$\therefore$ A is the correct option.

28.) SOLUTION (D)

$$y=[3245.7]$$

$$=[3.2457 * 1000]$$

$$=[3.2457 + 1000]$$

$$=[\text{less than } 1 + 3]$$

$$=3$$

So, the answer comes out to be 3

$\therefore$ D is the correct option.

29.) SOLUTION (A)

$$1440 = 2^5 \times 3^2 \times 5^1$$

$$\text{No of factors of } 1440 = (5+1)(2+1)(1+1) = 36$$

1440 is not a perfect square so no of ways it can be written as product of two factors = $36/2 = 18$

If n is not a perfect square then no of ways it can be written as product of factors = $(\# \text{ of factors of } n)/2$

If n is a perfect square then no of ways it can be written as product of factors = $(\# \text{ of factors of } n - 1)/2$

30.) SOLUTION (D)

$$\text{Let } k=n=2$$

Now,

$$(kn)! = (2 \times 2)! = 4! = 24$$

which is not divisible by $2^{kn} = 2^4 = 16$

31.) SOLUTION (C)

For a number n in the series let $x = n(n+1)/2 \dots (1)$

Here, x is the last position of the element n in the sequence.

$$A/q \ n(n+1)/2 = 5,000$$

$$\Rightarrow n(n+1) = 10,000 ; \text{ which simply means } n \sim 100$$

Put $n=100$ in (1), $100(101)/2 = 5050$ i.e. position of last 100 is 5050 and position of first 100 must be $5050 - (100 - 1) = 4951$.

Therefore, from position 4951 to 5050 there are 100 100s, hence the correct answer is option C.

32.) SOLUTION(C)

$$3528 = 2^3 \times 3^2 \times 7^2$$

$$\text{Number of proper factors} = (3+1)(2+1)(2+1) - 2 (\text{for } 1 \text{ and } 3528) = 34$$

33.) SOLUTION (A)

Since the price per item is lower for mangoes, one can only buy mangoes here.

No. of mangoes one can get for Rs.1=10/100=0.1

No. of mangoes one can get for Rs.150=150×0.1=15

34.) SOLUTION: (A)

back and forth

35.) SOLUTION (A)

The line starts from P which is inside 575 contour line.

Then line crosses the contour lines 575→550→525→500. **Down**

Then it crosses the contour lines of 500→525→550→575 **Up**

Then it crosses the contour lines 575→550 and then stops at Q which is inside the contour line 525. **Down**

∴ Option C Down up Down is the correct answer.

36.) SOLUTION (C)

If we look at the equation it is $a^2+2ab+b^2$ where $2ab=11.98*a$
 $= (11.98)^2+2ab+(0.02)^2$

To make it a perfect square $b=0.02$ and $a=11.98$

$2*11.98*0.02=11.98*a$

$\Rightarrow a=0.04$

So the number now will be a perfect square of

12 ($\because (11.98+0.02)^2=12^2$)

∴ 0.04 is the correct answer.

37.) SOLUTION (D)

From the graph we can see that movements

between 1-2, 1-3, 1-5, 2-3 never happened. So, statement 1 is not TRUE.

Statement 2 is also FALSE as the elevator stays on fourth floor for 19 minutes while it stays at ground floor for 21 minutes.

38.) SOLUTION (D)

Let the speed by cycling be X km/h and speed of walking be Y km/h

So, $(1/2)X + (3/2)Y = 19 \rightarrow (1)$

And, $X + Y = 26 \rightarrow (2)$

Solving (i) & (ii) we get, $X = 20$ and $Y = 6$

Thus, the speed in km/h at which Budhan walk is 6.

39.) SOLUTION (C)

Using first letter to denote the cities, we can arrange them in the order of elevations as (, means order is not sure)

$\{ \{S < P\}, K \} < G < D$

So, options (ii) and (iv) can be concluded whereas we are not sure about (i) and (iii).

40.) SOLUTION (A)

"He saw that people inevitably came out of the centre in the order that they went in"

This means customers were serviced on FCFS basis. This rules out options B and C.

"However, the time they spent inside seemed to vary a lot"

This makes option A correct.

Option D in no way can be inferred from the given passage.

41.) SOLUTION(C)

Let's supposed added number be x Then, the equation will be

$$2 + x/5 + x = 5/6$$

$$\Rightarrow 12 + 6x = 25 + 5x$$

Required number $= x = 13$.

Hence, option C is correct.

42.) SOLUTION (D)

Greatest six-digit number is 999999.

Divide this number by 12 and get remainder as 3.

Since the remainder is 3, if you subtract 3 from the number, the

remaining number will be a multiple of 12.
So the greatest such number will be $999999-3=999996$.

43.) SOLUTION (D)

Ans d average

44.) SOLUTION (B)

An axiom is a statement or proposition which is regarded as being established, accepted, or self-evidently true.

45.) SOLUTION (C)

Population Unemployment Poverty Disease Death

24153

46.) SOLUTION (B)

AB= length of thread =180 meter

$\angle PQR=60^\circ$

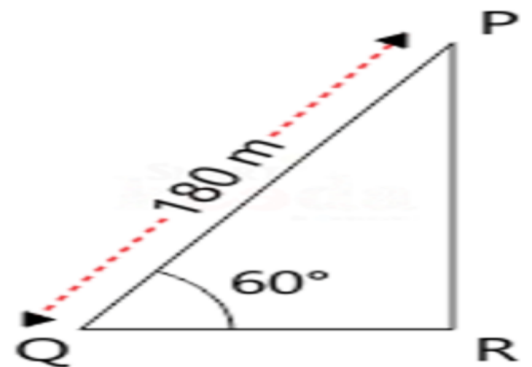
In $\triangle PQR$

$$\sin 60^\circ = \frac{PR}{PQ} = \frac{\sqrt{3}}{2} = \frac{PR}{180}$$

$$\text{so, } \frac{PR}{180} = \frac{\sqrt{3}}{2}$$

$$PR = 180 \times \frac{\sqrt{3}}{2} = 90\sqrt{3}$$

Hence, option B is correct



47.) SOLUTION (B)

Let the article costs 'X' to A

- Cost price for B = $1.2X$
- Cost price for C = $0.75(1.2X) = 0.9X$
- Cost price for D = $1.4(0.9X) = 1.26X = 252X = 200$

Amount paid by A for the article = Rs. 200.

48.) SOLUTION (A)

Let the first number be x and the second number be y.

According to the question,

60% of x = 40% of y $\Rightarrow x/y = 40/60$

So, the required ratio = x:y = 2:3.

Hence, option A is correct.

49.) Solution: (D)

The correct answer is option (d) as it says that the exam is not GATE, the exam is PGEE which means the same as the given statement

50.) Solution: (A)

3 or 4 or 5 or 6 are the possible outcomes. Hence $4/6 = 2/3$ will be the required probability.

51.) SOLUTION C.

linear time for enqueue and constant time for dequeue

It is given that we are maintaining a reference to the item least recently added (the front of the list) but not maintaining a reference to the item most recently added (the end of the list). So we are sure that the head of the linked list = front of the list = least recently added = added at the beginning = to be dequeued at first (since it is a Queue and Queue uses FIFO policy).

We know that it takes constant time to remove a node from the front of a linked list. (Make temp point to head, move head to next node, delete temp). Therefore, Dequeue will take constant time.

We are also sure that, the end of the Null-terminated linked list = last element of the list = most recently added = added at the end = the place where next enqueue must be done (since it is a Queue and Queue uses FIFO policy).

However, we know that it takes linear time to find the last node of a linked list. (We need to traverse till end of LL as we are not maintaining a reference to it). Therefore, Enqueue will take linear time.

52.) Solution: (B)

Given that attribute B is a part of attribute set X and it is not prime says that a non-prime attribute is determining another attribute.

We know that, given a FD $X \rightarrow Y$ will be known as a transitive dependency if both X and Y are non-prime attributes. And the relation having a transitive dependency will not be in 3NF. Therefore, the relation R is in 2NF but might not be in 3NF.

53.) SOLUTION D

Since, G and F are disjoint, they cannot happen together.

$P(G|F)$ means Probability of event G happening given event F happened.

Since F has happened and they both cannot happen together, we know G cannot happen.

So, $P(G|F)=0$

54.) SOLUTION A

Let's take a set $A=\{a,b\}$

its powerset will have 4 elements

make another set B having all these elements and some more elements(lets say total =5) . Then option B,C is false.

Let's take a set $A=\{a\}$

its powerset will have 2 elements

make another set B having both of these elements. Now $|A|+2=3$ and $|B|=2$, so Option D is false.

55.) SOLUTION d

Bottom of Form
Top of Form
Bottom of Form
Top of Form

$$a \oplus b = a' \odot b; \text{ So, } R-3$$

56.) Solution c

We have a state of deadlock if and only if the wait-for graph has a cycle.

57.) SOLUTION B

Let n be the number of pages visited by the search engine at the time a query is submitted. Assume that it takes constant time to compute the relevance score for each page w.r.t. a query. Then it takes $O(n)$ time to compute the relevance scores, a further $O(n)$ time to build a heap of n relevance scores, and $O(k \log n)$ time for k delete-max operations to return the top k scores.

58.)Solution D

If values are stored in an array then

- for ascending order it just takes $\theta(1)$ because the very first value is always first negative value.
- for descending order it takes $\theta(\log n)$ because we can apply binary search to find first negative.

If values are stored in a linked list then

- for ascending order it just takes $\theta(1)$ because the very first value is always first negative value.
- for descending order it takes $\theta(n)$ because we can not apply binary search to find first negative.

59.) SOLUTION B

No of nodes at level 1 of tree $\Rightarrow 2$

No of nodes at level 2 of tree $\Rightarrow 4$

No of nodes at level 3 of tree $\Rightarrow 8$

No of nodes at level 4 of tree $\Rightarrow 16$

Last node in level 4th is the node we are looking for $\Rightarrow 1+2+4+8+16 \Rightarrow 31$

60.) SOLUTION C

Since $g(x)$ is a polynomial function it is continuous and differentiable everywhere. Therefore there are two theorems that we might use to answer this question. The first is the Intermediate Value Theorem, which says that between -1 and 2 and any y -value between 4 and 7 there is at least one number c such that $g(c)$ is equal to that y -value. Since none of the answer choices involve y -values between 4 and 7 , we go on to the next theorem.

The Mean Value Theorem says that between -1 and 2 there is at least one number c such that $g'(c)$ is equal to the slope of the secant line between the points $(-1, 4)$ and $(2, 7)$, i.e. 1 . So C is correct.

61.) SOLUTION A

Differentiate $f(x)$ we get $100x^{99} + 99x^{98} + 98x^{97} + \dots + 1$. Put $x=1$ then it is just sum from 1 to 100 . So $(100 \cdot 101)/2 = 5050$

62.) SOLUTION D

$$F(A, B, C, D) = A'B'C + A'B'D + AB'C' + ABC'D'$$

So, Option D is the Correct choice.

63.) SOLUTION A

The I/O-bound tasks incur a context switch after using up only 1 millisecond of the time quantum. The time required to cycle through all the processes is, therefore, $(10 \cdot 1.1) + (10.1)$ (as

each I/O-bound task executes for 1 millisecond and then incurs the context switch task so($10*(1+0.1)$), whereas the CPU-bound task executes for 10 milliseconds before incurring a context switch($10+0,1$)).

So total time = cpu bound + io bound

Cpu bound= $10+0.1$ (only one time context switch)= 10.1

Io bound = $10(1+0.1)=10*(1.1)=11$

Useful time = 20 .

Total time = $10.1+11=21.1$

The CPU utilization is, therefore, $(20/(21.1))*100=94\%$.

64.) SOLUTION A

The receiver can correctly decode only if the received codeword is having utmost one-bit flip. That is either no bit is flipped or just one bit is flipped.

$P(\text{decode correctly}) = P(\text{no bit flipped}) + P(\text{exactly one bit flipped})$

$P = (1-p)^7 + 7p(1-p)^6$.

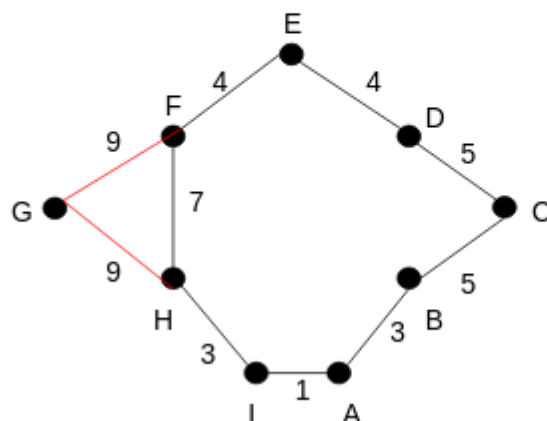
$= 0.9^7 + 7 \times 0.9^6 \times 0.1 = 0.8503$.

65.) SOLUTION A

(A) A,B - Enqueue C,D – Dequeue

66.) Solution: (A)

If we apply Kruskal's algorithm we will end up getting the following graph with two choices among GH and GF, hence 2 MST's are possible



67.) Solution: (B)

In selective repeat,

Sequence number is 1024 , so m should be 10 bits

$$\text{SWS} \leq 2^{(m-1)}$$

The max send window size = 512

$$T_t = 100 * 8 \text{ bits} / 100 * 10^6 = .008 \text{ milli sec}$$

$$T_p = 20 \text{ millisec}$$

$$a = T_p / T_t = 20 / .008 = 2500$$

$$\text{Eff} = WS / 1 + 2a$$

$$= 512 / 1 + 2a$$

$$= 512 / 5001 = 10.237\%$$

$$\text{Throughput} = \text{Eff} * \text{bandwidth}$$

$$= 10.237/100 * 100 \text{ Mbps} = 10.23 \text{ Mbps} = 10(\text{Ans})$$

68.) Solution: A

$$\text{Sun} = 0$$

$$\text{Mon} = -6$$

$$\text{Tue} = -5$$

$$\text{Wed} = -4$$

$$\text{Thu} = 4$$

$$\text{Fri} = 8$$

$$\text{Sat} = 9$$

$$\text{Sun} + \text{Mon} + \text{Tue} + \text{Wed} + \text{Thu} + \text{Fri} + \text{Sat} = 6$$

69.) Solution: (D)

$$\text{The number of blocks} = 112 \text{ GB} / 2 \text{ KB} = 56 \text{ M blocks}$$

For each block, bitmap needs 1 bit to keep track.

$$\text{So, we will have, } 56 \text{ M blocks} * 1 \text{ bit} = 56 \text{ Mb}$$

$$\text{Therefore, we need } 56/8 = 7 \text{ MB}$$

70.) Solution A

71.) SOLUTION C

$P(\text{Good club}) = 1/5$

$P(\text{Bad club}) = 1 - P(\text{good club}) = 1 - 1/5 = 4/5$

ITEM	PROFIT	WEIGHT	PROFIT/WEIGHT
1	5	1	5
2	6	2	3
3	4	2	2
4	8	1	8

$P(\text{Good Shot}) = P(\text{Good club}) * P(\text{good shot/good club}) + P(\text{good shot/bad club})$

$= 1/5 * 1/3 + 4/5 * 1/4 = 16/60 = 4/15$.

72.) SOLUTION B

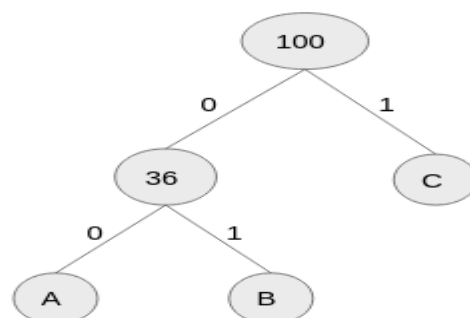
For each subject the student may either pass or fail so there are 2 ways in all there are 26 ways for the 6 subjects, out of which there is only one way to pass hence $26 - 1 = 63$ ways to fail.

73.) SOLUTION A

The optimal solution would be to include items 4, 1, 3, and 2
 $5 + 6 + 4 + 8 = 23$

74.) Solution B

One of the possible Huffman tree is as follows



In the new text we have 4 a and 6 bs and one cs the length of the Huffman code is $4 * 2 + 6 * 2 + 1 = 21$

75.) Solution C

$$\begin{aligned} & \int_0^{\pi} \sin x \cos^4 x \, dx \\ &= - \int_0^{\pi} (-\sin x) \cos^4 x \, dx \\ &= [\cos^5 x / 5]_0^{\pi} \\ &= 2/5 \end{aligned}$$

76.) Solution c

We know that the closure of a primary key or a candidate key in a relation R consists of all the attributes of relation R. Therefore an alternative method to establish that a set of attribute X forms a key of relation R is by checking if R is equal to the closure of the set X

77.) Solution: (D)

The worst case of quick sort corresponds to the case when the pivot divides the array into subarrays of the form, 0 and n-1.

The options

9, 1, 0, 2, 8, 4, 7, 6, 5, 3--this corresponds to the sequence, 10, 11, 12, 13, 17, 21, 24, 25, 30, 45.

3, 5, 6, 7, 4, 8, 2, 0, 1, 9--this corresponds to the sequence, 45, 30, 25, 24, 21, 17, 13, 12, 11, 10.

3, 9, 5, 1, 6, 0, 7, 2, 4, 8- this corresponds to the sequence, 45, 10, 30, 11, 25, 12, 24, 13, 21, 17.

Each of the above sequences corresponds to n-1 and 0 split in each case.

9, 1, 8, 6, 3, 0, 2, 5, 7, 4 - this corresponds to the sequence, 10, 11, 17, 25, 45, 12, 13, 30, 24, 21. This does not correspond to 0, n-1 split at each

78.)Solution A

$$S = \{a, b, c\}, = \{1, 2, 3\}$$

(i) $F: S \rightarrow T$ is defined as $F = \{(a, 3), (b, 2), (c, 1)\}$

So, $F(a) = 3$, $F(b) = 2$, $F(c) = 1$

Therefore, $F^{-1}: T \rightarrow S$ is given by $F^{-1} = \{(3, a), (2, b), (1, c)\}$.

(ii) $F: S \rightarrow T$ is defined as $F = \{(a, 2), (b, 1), (c, 1)\}$ Since $F(b) = F(c) = 1$,

F is not one - one.

Hence, F is not invertible i.e., F^{-1} does not exist

79.) Solution: $\rightarrow D$

The Domain of f (D_f) = $\{2, 5, 8, 10\}$ and Domain of g (D_g) = $\{2, 7, 8, 10, 11\}$.

By algebra of functions, the sum $f + g$ is the function whose domain is the set of all numbers common to the domains of f and g .

Therefore domain of $f + g = \{x \mid x \text{ belongs to } D_f \cap D_g\} = \{2, 8, 10\}$

80.) SOLUTION C

It is given that $P(\sim A):P(A)=3:2$ $P(A)=2/5$

$P(\sim B):P(B)=5:4$ $P(B)=5/9$

The event that A solves the problem is independent from the event that B solve the problem

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 2/5 + 5/9 - (2/5) * (5/9) = 11/15$

81.) Solution: (D)

If we solve for LCS using the dynamic programming table we will get 3 strings which form the longest common subsequence which are of length 7 and are as follows.

ahahaah

ahahaha
hahaaha

82.) Solution B

Order of the following execution is non deterministic as parent and child run concurrently

83.) Solution: (B)

In this method, the first job claims the first available memory with space more than or equal to its size. The operating system doesn't search for an appropriate partition but just allocate the job to the nearest memory partition available with sufficient size.

First-Fit takes 20KB(remaining 8KB) for request of 12 KB,

Then takes 10KB for request of 10KB, and

Finally(for third segment request) takes 20KB for a request of 8 KB.

84.,.) Solution: (D)

1. Both have the checksum of the 16 bits.
2. UDP does not provide flow as well as the congestion control, TCP have both.
3. TCP is always connection oriented.
4. Yes

85.) Solution: (C)

1. DHCP is used for this purpose.
2. Port number of DHCP is 67.
3. True
4. Only, SMTP is a push protocol.

86.) Solution: (B)

Each circuit has a transmission rate of $(1.536 \text{ Mbps})/24 = 64 \text{ kbps}$

Time to transmit a file = $(640,000 \text{ bits})/(64 \text{ kbps}) = 10$

Total time = 10 seconds + 500 ms (circuit establishment time) = 10

.5 sec

87.) Solution A

We require 2 RTT for transmitting every object in the non persistent HTTP

Sp base file +6 object =7

Sp 7×2 RTT =14 RTT required

Sp $14 \times 20 = 280$ m sec

88.) SOLUTION B

No of ways we can take one or more at a time is

$$C(n,1) + C(n,2) + C(n,3) + C(n,4) + \dots + C(n,n) = 2^n - 1. \quad 2^n - 1 = 127$$

$$2^n = 128 \quad n = 7.$$

89.) Solution: (D)

Need :

$$P1 = \text{Max - Allocation} = 2 \quad 1 \quad 0 \quad - \quad 0 \quad 1 \quad 0 = 2 \quad 0 \quad 0$$

$$P4 = 3 \quad 2 \quad 1 \quad - \quad 2 \quad 1 \quad 1 = 1 \quad 1 \quad 0$$

Since available is $2 \quad 0 \quad 0$, therefore, P1 and P4 are able to finish.

90.) Solution D

$$T_p = 20 \text{ msec}$$

$$a = T_p / T_t$$

$$T_t = \text{frame size} / \text{Bandwidth}$$

$$\text{Eff} = 1 / (1 + 2a) = 0.5 = T_t / (T_t + 2T_p)$$

$$0.5 T_t + T_p = T_t$$

$$0.5 T_t = T_p$$

$$T_t = 20 \text{ msec} / 0.5 = 40 \text{ msec}$$

$$\text{Frame size} = \text{BW} \times 40 \text{ msec} = 4 \times 10^3 \times 40 \times 10^{-3} = 160 \text{ bits}$$

91.) Solution: (A)

In order to construct a balanced BST the root is initially chosen as the middle element and the left child of the root is the middle element.

92.) Solution: D

$$\begin{aligned} F(P, Q, R, S, T, U) &= PRT + QS'T' + PST(R + T'U) (QR' + SU) \\ \Rightarrow PRT + QS'T' + PSTR + PSTT'U (QR' + SU) \\ \Rightarrow PRT + QS'T' + PSTR (QR' + SU) \quad //PSTT'U \text{ will be 0 due to } TT' \\ \Rightarrow PRT + QS'T' + PSTRQR' + PSTRSU \\ \Rightarrow PRT + QS'T' + PSTRU \quad //PSTQRR' \text{ will be 0 due to } RR' \\ \Rightarrow PRT(1 + SU) + QS'T' \\ \Rightarrow PRT + QS'T' \quad // (1 + SU) = 1 \\ \Rightarrow PRT + QS'T' \end{aligned}$$

Therefore, the number of literals will be 6.

93.) SOLUTION C

Note consider the preemptive priority scheduling

P0	P1	P0	P2
0	1	5	
7			

WAITING TIME = $0+5-1=4$ unit

94.) Solution C

95.) SOLUTION B

a is the array

"ECENT" is the string constant

ptr=a// ptr will start pointing to the array a.

strcpy(a, ptr)// both pointer pointing to the same array so DECENT is replaced by DECENT

*ptr = D +1 // ascii

96.) Solution: (A)

`const char *a` // here-string is constant but the pointer is not constant.

So `a = string` // a start pointing to R.

`while(*a)` // will fetch the array values, the loop will fail when NULL will come

In the loop it print ROUGH.

97.) SOLUTION D

$\text{foo}(\text{arr}, 5) = *(\text{arr}) + \text{foo}(\text{arr}+1, n-1) = 10 + \text{foo}(\text{arr}+1, 4)$

$\text{foo}(\text{arr}+1, 4) = *(\text{arr}+1) - \text{foo}(\text{arr}+1, n-1) = 30 - \text{foo}(\text{arr}+2, 3)$

$\text{foo}(\text{arr}+2, 3) = *(\text{arr}) + \text{foo}(\text{arr}+1, n-1) = 30 + \text{foo}(\text{arr}+3, 2)$

$\text{foo}(\text{arr}+3, 2) = *(\text{arr}+1) - \text{foo}(\text{arr}+1, n-1) = 50 - \text{foo}(\text{arr}+4, 1)$

$\text{foo}(\text{arr}+4, 1) = *(\text{arr}) + \text{foo}(\text{arr}+1, n-1) = 50 + \text{foo}(\text{arr}+5, 0)$
 $\text{foo}(\text{arr}+5, 0) = 0$

So finally $= \text{foo}(\text{arr}, 5) = 10 + 30 - 30 + 50 - 50 + 0 = 10$

98.) SOLUTION B

First we can select one in lhs rest of the element in rhs

so $4c1*(3c1+3c2+3c3)+4c2*(2c1+2c2)+4c3*(1c1)$

sum is 50

99.) solution D

Division Operation Notation: r/s or $r \div s$

Useful for expressing queries that include a "for all" or "for every" phrase.

100.) Solution A

A. The weak entity set must participate totally in the identify weak relationship.

Usually, there exist one to many relationship between strong entity and weak entity thus total participation from weak entity set.

